Ticket 7

TCK07-DnsFallback

Topics: DNS, NetBIOS/LLMNR/mDNS name resolution fallback, WINS, DNS suffix and zone matching, DNS cache TTL

1. Summary

Title	User can't resolve server by NetBIOS name — DNS adapter misconfigured
Date	2026-10-31
Ticket ID	TCK07-DnsFallback
Status	Resolved

2. Problem

• **Symptom:** User (acarter) reported being unable to open shared drives or internal sites; Outlook and general internet worked fine. (Simulated: DNS server on the domain-facing adapter was pointed to 8.8.8.8 instead of the DC.)

• **Impact:** Access to the Sales share and other internal/domain resources.

• **Started when:** After the break script was administered on the client VM.

• **Recent changes:** DNS server address on the domain-facing NIC (Ethernet 2) changed from the DC's IP (192.168.50.10) to 8.8.8.8.

3. Environment

• **Device / OS:** Client VM, Windows, joined to acme.local

• **Network / Server:** Domain Controller WIN-4G8SQK9BMC1 (192.168.50.10), AD lab

• **Software / Version:** PowerShell

4. Troubleshooting Steps

#	What I tried	Result
1	Ran the break script by copy-pasting the decoded contents directly into an admin PowerShell prompt	Didn't work
2	Navigated to the folder containing the .ps1 file and ran it from there	Blocked — "running scripts is disabled on this system"
3	Ran `Set-ExecutionPolicy -ExecutionPolicy Bypass -Scope Process`, then re-ran the script	Script executed
4	Checked `\\WIN-4G8SQK9BMC1\Sales` — still accessible despite the break	Initially assumed break hadn't landed
5	Ran `Get-DnsClientServerAddress -AddressFamily IPv4`	Confirmed break had landed on Ethernet 2 (the domain-facing adapter), set to 8.8.8.8
6	Ran `Resolve-DnsName WIN-4G8SQK9BMC1`	Resolved successfully via `.local` suffix — a fallback mechanism, not real DNS, was masking the break
7	Ran `Resolve-DnsName WIN-4G8SQK9BMC1 -Type A -DnsOnly`	Timed out with a DNS failure exception, confirming DNS itself was broken
8	Reverted DNS server to the DC's IP	—
9	Ran `Resolve-DnsName WIN-4G8SQK9BMC1` again	Resolved via `acme.local`, TTL 3600
10	Ran `Resolve-DnsName WIN-4G8SQK9BMC1 -Type A -DnsOnly` again	Returned "DNS server failure" — a separate, pre-existing issue unrelated to the break

5. Root Cause

The domain-facing adapter's DNS server was pointed to 8.8.8.8 instead of the DC, breaking real DNS resolution. Access still appeared to work because an unqualified NetBIOS-style name query fell back to a link-local multicast mechanism (not the configured DNS server) to resolve the host.

6. Fix

Reverted the DNS server address on the domain-facing adapter (Ethernet 2) back to the DC's IP.

```
Set-DnsClientServerAddress -InterfaceAlias "Ethernet 2" -ServerAddresses '192.168.50.10'
```

Commands used during troubleshooting:

```
Get-DnsClientServerAddress -AddressFamily IPv4
Resolve-DnsName WIN-4G8SQK9BMC1
Resolve-DnsName WIN-4G8SQK9BMC1 -Type A -DnsOnly
Resolve-DnsName WIN-4G8SQK9BMC1.acme.local -DnsOnly
Clear-DnsClientCache
```

7. Verification

After reverting DNS, `Resolve-DnsName WIN-4G8SQK9BMC1` returned a fresh `acme.local` record with TTL 3600 (a full hour, confirming it was a live, just-cached lookup rather than stale data). Share access was confirmed as working through correct DNS rather than the fallback mechanism.

8. Lessons Learned

• **What I'd check first next time:** Don't assume access working means the break didn't land — check which adapter is actually in use, and test with `DnsOnly` to rule out fallback mechanisms masking a real DNS failure.

• **What I got wrong or assumed:** Initially assumed the break script failed because the Sales share was still reachable. It was actually working as intended — a fallback resolution mechanism (NetBIOS/LLMNR/mDNS-style, learned as WINS: the legacy NetBIOS-name-to-IP database) was resolving the name instead of DNS.

• **How to prevent it:** Understand that unqualified hostname lookups get a DNS suffix appended by the resolver (e.g. `acme.local`) so the DNS server can match the query to a hosted zone; without that suffix, the server has no matching zone to search. Also remember DNS records cache with a TTL (default 3600s/1hr) — a record found after that window has expired is not from cache, and one found within it may be.

9. Screenshots

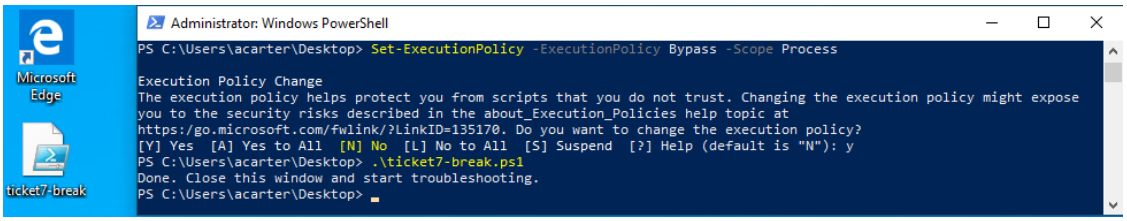


Figure 1. Execution policy set to Bypass for the session, then ticket7-break.ps1 run and reports Done

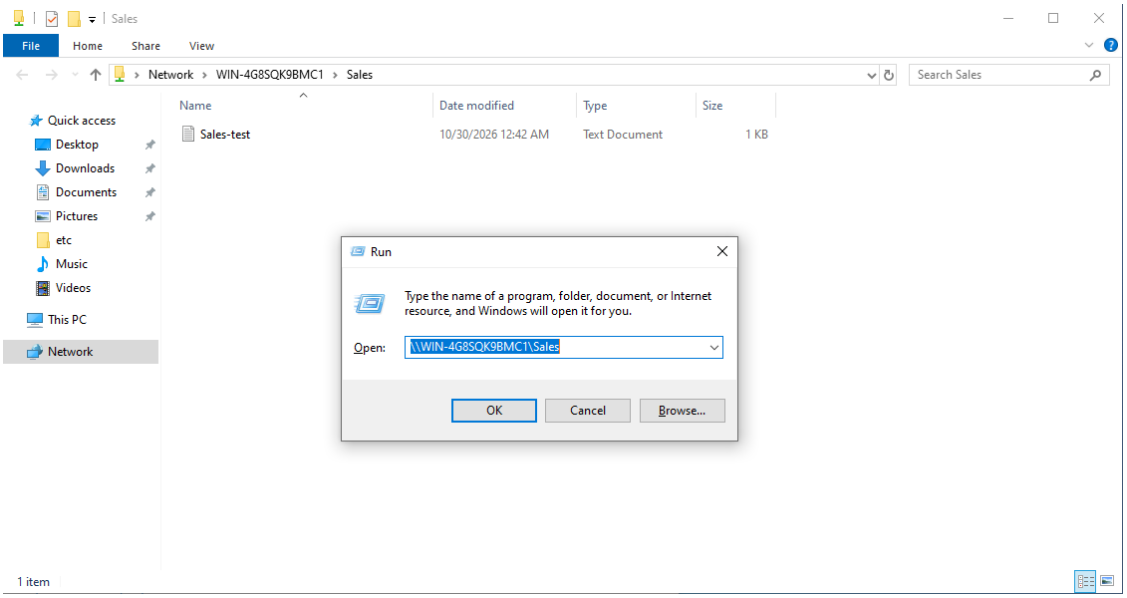


Figure 2. Sales share still opens through \\WIN-4G8SQK9BMC1\Sales after the break

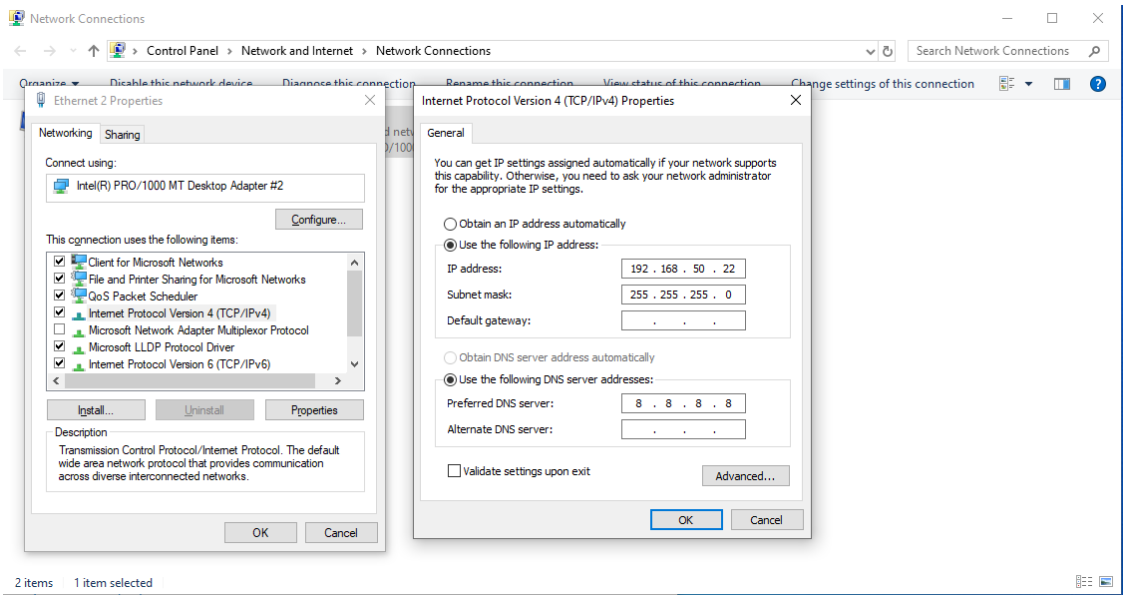


Figure 3. Ethernet 2 IPv4 properties with the preferred DNS server set to 8.8.8.8

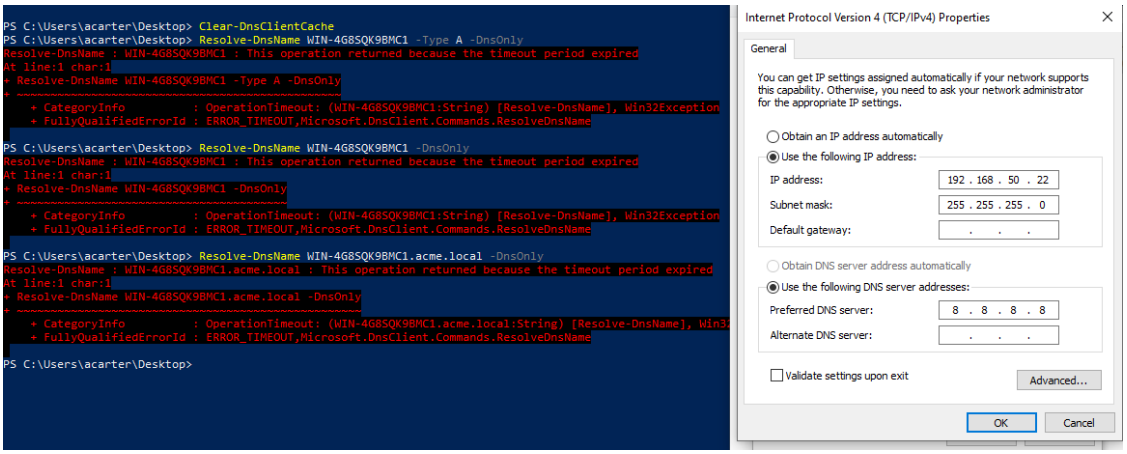


Figure 4. Resolve-DnsName with -DnsOnly timing out while DNS is set to 8.8.8.8

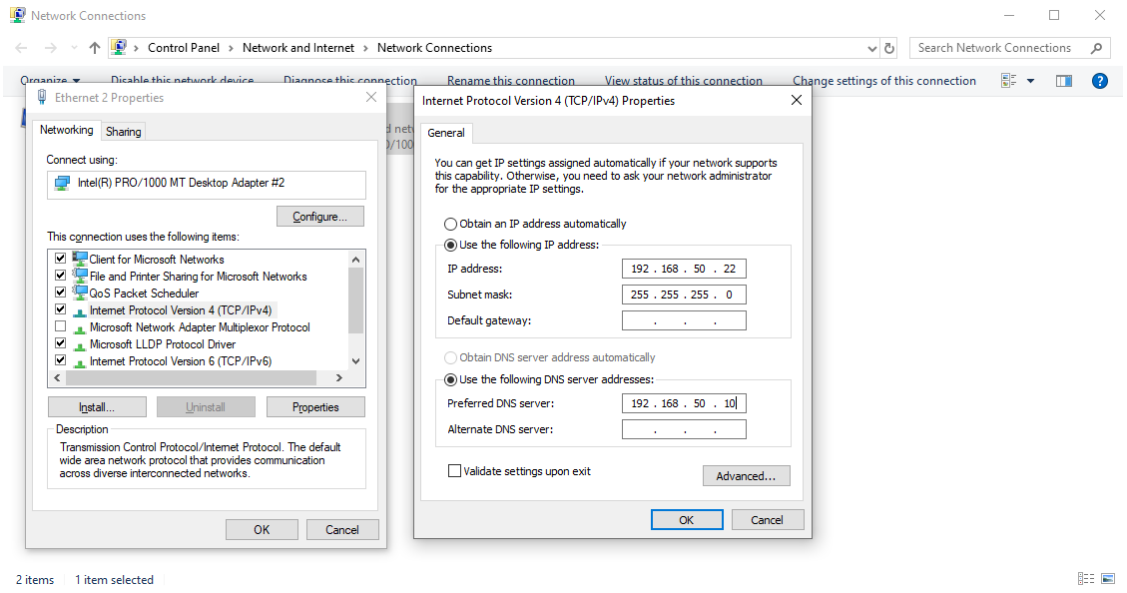


Figure 5. Ethernet 2 IPv4 properties with the preferred DNS server reverted to 192.168.50.10

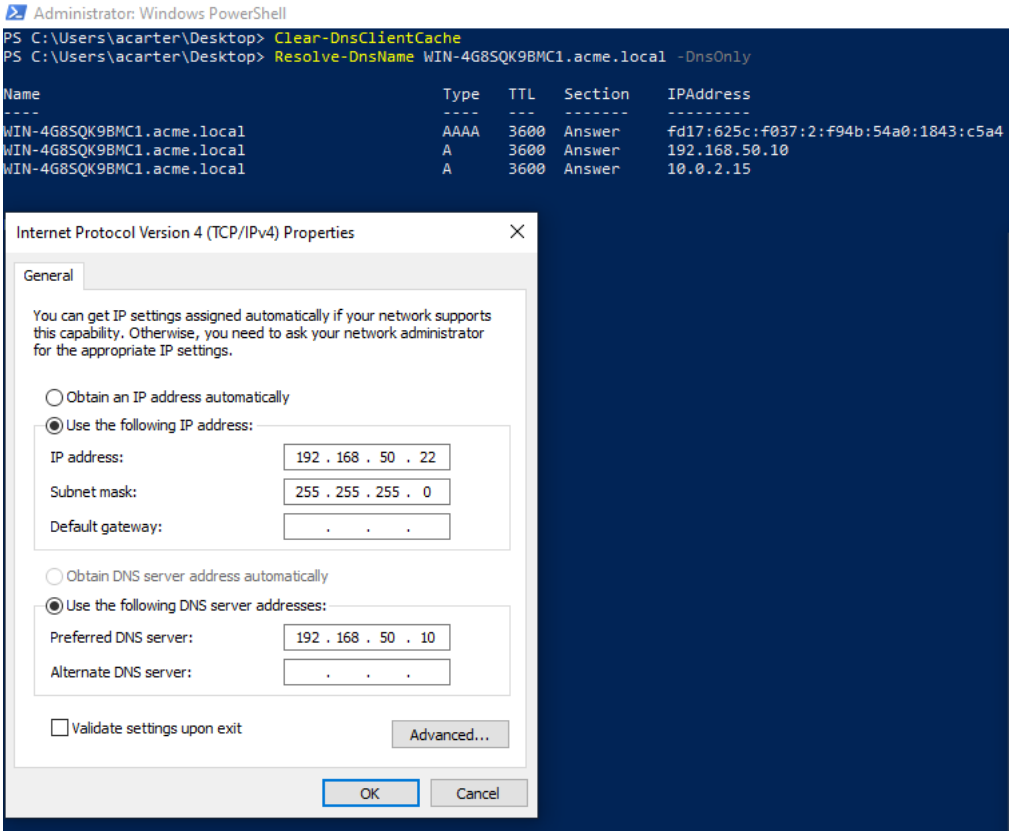


Figure 6. Resolve-DnsName WIN-4G8SQK9BMC1.acme.local -DnsOnly resolving with TTL 3600 after the revert